

Chapter 08: Regression Analysis

Regression analysis is the foundation of predictive modeling in AI. It maps the relationship between a dependent variable (Y) and independent variables (X). In Data Science, it's used for forecasting (e.g., sales, stock prices) and inferring causal relationships.

Types of Regression

- Simple Linear Regression: One predictor variable
- Multiple Linear Regression: Multiple predictor variables
- Non-linear Regression: Non-linear relationships between variables

8.1 Simple Linear Regression (SLR)

The simplest deterministic mathematical relationship between two variables x and y is a linear relationship $y = \beta_0 + \beta_1 x$. The set of pairs (x, y) for which $y = \beta_0 + \beta_1 x$ determines a straight line with slope β_1 and y -intercept β_0 .

More generally, the variable whose value is fixed by the experimenter will be denoted by x and will be called the independent, predictor, or explanatory variable. For fixed x , the second variable will be random; we denote this random variable and its observed value by Y and y , respectively, and refer to it as the dependent or response variable.

Usually observations will be made for a number of settings of the independent variable. Let $x_1, x_2, \dots, x_n$ denote values of the independent variable for which observations are made, and let Y_i and y_i , respectively, denote the random variable and observed value associated with x_i . The available bivariate data then consists of the n pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. A picture of this data called a scatterplot gives preliminary impressions about the nature of any relationship. In such a plot, each (x_i, y_i) is represented as a point plotted on a two-dimensional coordinate system.

Example 8.1:

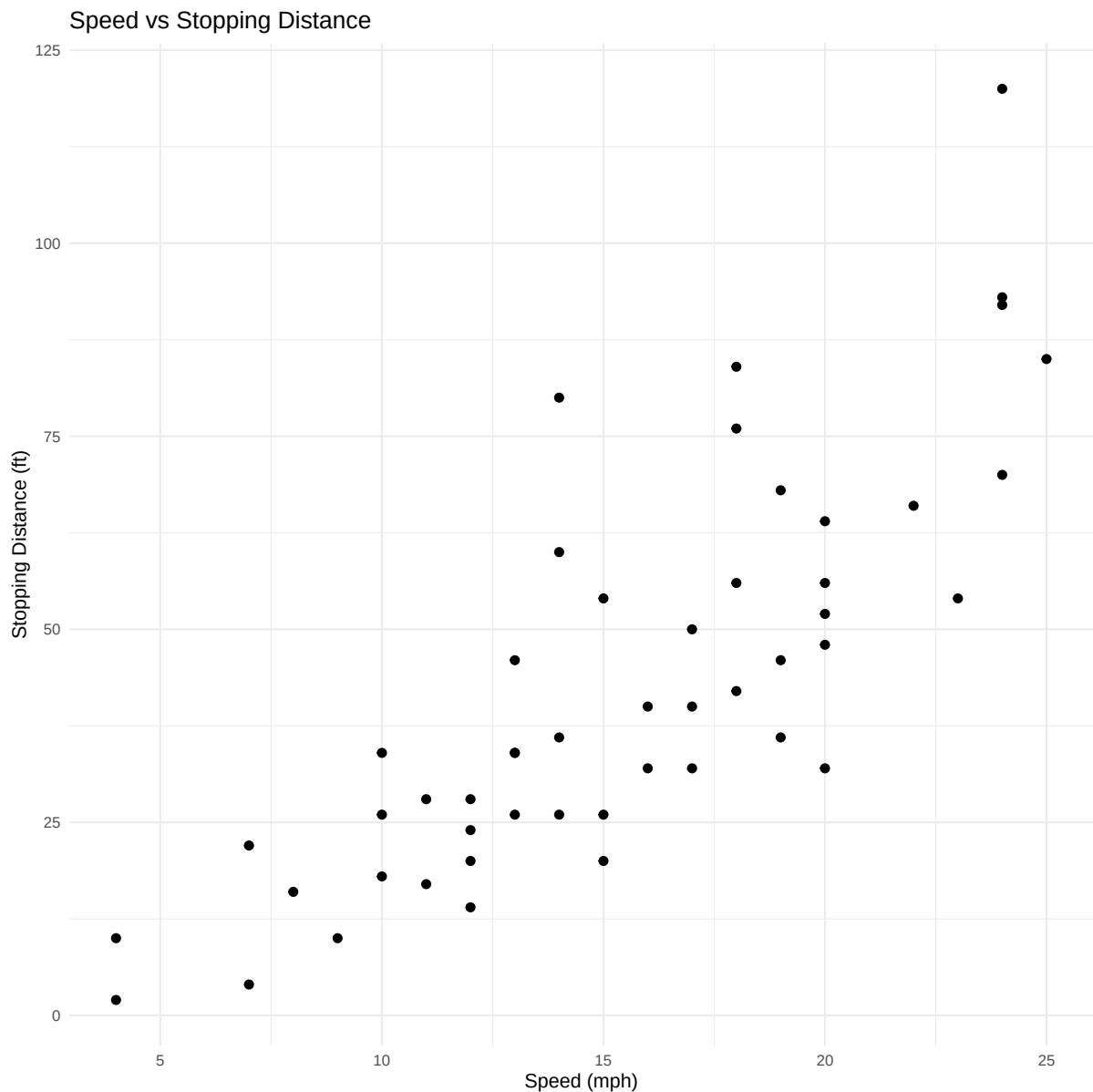
The built-in dataset `cars` contains observations on the speed of a car and the corresponding stopping distance. Suppose a traffic safety researcher is interested in understanding how the stopping distance of a car depends on its speed.

Let

X = Speed of the car (mph), Y = Stopping distance (ft).

1. Calculate the correlation coefficient and draw a scatter plot of stopping distance against speed. Comment on the nature of the relationship between X and Y .

```
ggplot(cars, aes(x = speed, y = dist)) +  
  geom_point() +  
  labs(title = "Speed vs Stopping Distance",  
        x = "Speed (mph)",  
        y = "Stopping Distance (ft)") +  
  theme_minimal(base_size = 9)
```



```
cor(cars$speed, cars$dist)
```

```
## [1] 0.8068949
```

The scatter plot and the correlation coefficient show a strong positive relationship between speed and stopping distance. As the speed of the car increases, the stopping distance tends to increase, suggesting a linear trend.

2. Assume that the relationship between speed and stopping distance can be modeled using a simple linear regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where ε is a random error term with mean zero and constant variance.

- (a) Explain the practical interpretation of the parameters β_0 and β_1 in this context.

- β_0 : the expected stopping distance when speed is zero (not physically meaningful, but part of the model).
- β_1 : the expected increase in stopping distance (in feet) for a one mph increase in speed.

- (b) State the key assumptions of the simple linear regression model.

to provide valid inference and reliable predictions, the following assumptions about the error term ε must hold.

- *Linearity*: The expected value of the response variable Y is a linear function of the predictor X , that is

$$\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 X.$$

This means that the change in the mean stopping distance is proportional to the change in speed. In practice, this assumption is assessed using a scatter plot of Y versus X or a residuals versus fitted values plot. A systematic curved pattern would indicate a violation of linearity.

- *Independence of errors*: The error terms $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ are assumed to be independent of each other. This implies that the stopping distance of one car does not influence the stopping distance of another. Independence is particularly important in data collected over time or space; violations often occur in time-series data or clustered observations.
- *Constant variance (Homoscedasticity)*: The variance of the error term is constant for all values of X , that is

$$\text{Var}(\varepsilon \mid X) = \sigma^2.$$

In this context, it means that the variability in stopping distance is roughly the same at low and high speeds. If the spread of residuals increases or decreases with fitted values, the data are said to be heteroscedastic, which can lead to inefficient estimates and invalid standard errors.

- *Normality of errors*: The error terms are assumed to be normally distributed:

$$\varepsilon \sim N(0, \sigma^2).$$

This assumption is not required for estimating the regression coefficients, but it is essential for hypothesis testing and constructing confidence intervals. Normality is commonly checked using a normal Q–Q plot of the residuals. Moderate departures from normality are usually acceptable for large samples.

- (c) Fit the simple linear regression model using the given dataset.

```
model <- lm(dist ~ speed, data = cars)
summary(model)

##
## Call:
## lm(formula = dist ~ speed, data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.069  -9.525  -2.272   9.215  43.201
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.5791     6.7584  -2.601   0.0123 *
## speed        3.9324     0.4155   9.464 1.49e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.38 on 48 degrees of freedom
## Multiple R-squared:  0.6511, Adjusted R-squared:  0.6438
## F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

- i. Write down the estimated regression equation.

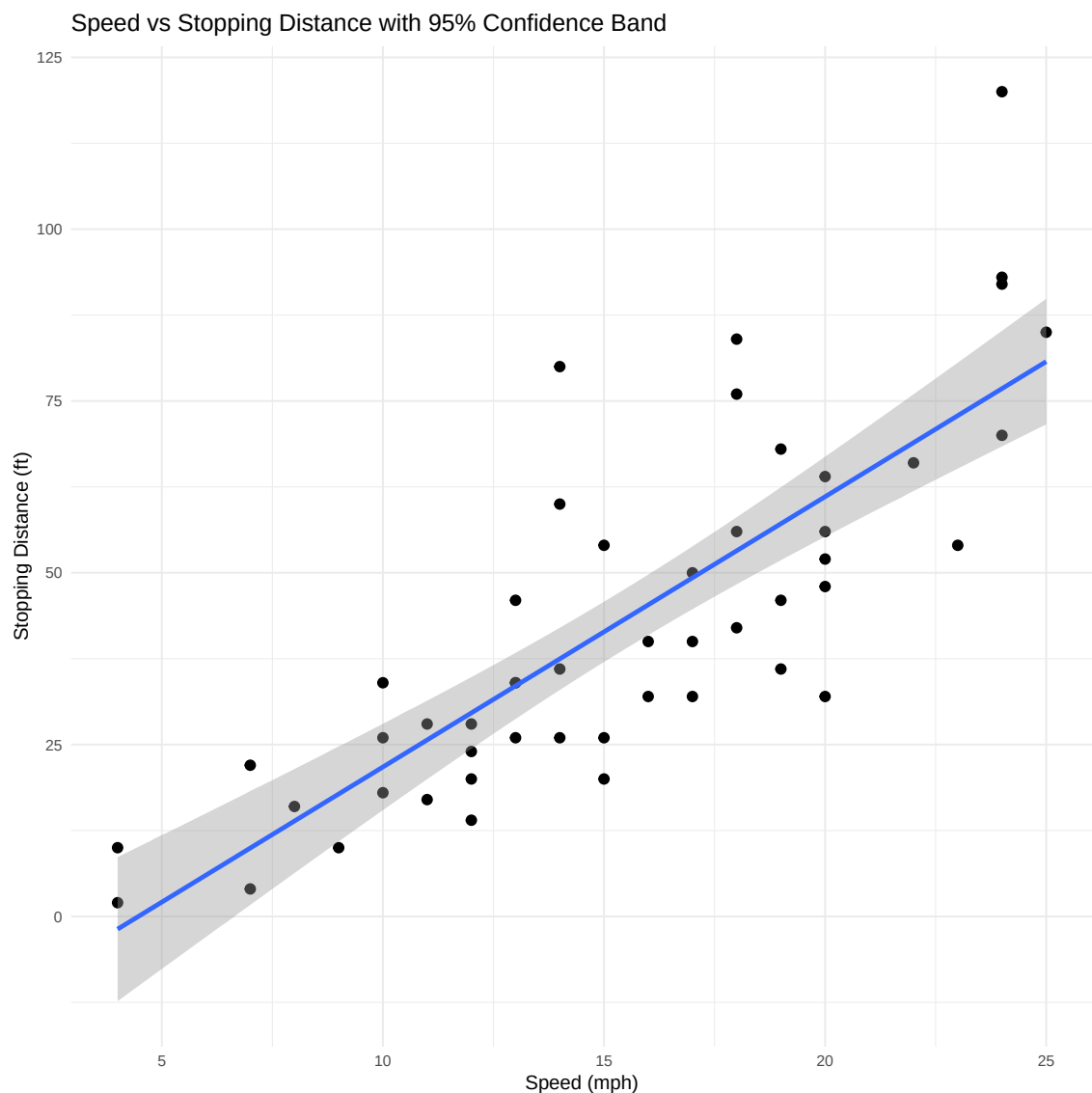
$$\hat{Y} = -17.58 + 3.93X$$

- ii. Interpret the estimated slope coefficient.

Interpretation of the slope: for each additional 1 mph increase in speed, the stopping distance increases on average by about 3.93 feet.

```
ggplot(cars, aes(x = speed, y = dist)) +
  geom_point(size = 1.8) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Speed vs Stopping Distance with 95% Confidence Band",
    x = "Speed (mph)",
    y = "Stopping Distance (ft)"
  ) +
  theme_minimal(base_size = 9)

## 'geom_smooth()' using formula = 'y ~ x'
```



- (d) Using the fitted model, predict the stopping distance for a car traveling at 50 mph.

```
new_data <- data.frame(speed = 50)
predict(model, new_data)

##          1
## 179.0413
```

- (e) Construct a 95% prediction interval for the mean stopping distance when the speed is 50 mph, and clearly explain its interpretation.

```
predict(model, new_data, interval = "confidence", level = 0.95)

##          fit          lwr          upr
## 1 179.0413 149.806 208.2766
```

A typical 95% prediction interval is approximately

$$\hat{Y} = 179.04, \quad 95\% \text{ PI} = (149.81, 208.28).$$

We are 95% confident that the stopping distance of a *single* car traveling at 50 mph will lie between approximately 150 ft and 208 ft.

- (f) What are the Null (H_0) and Alternative (H_1) hypotheses for testing the significance of the slope?

$H_0 : \beta_1 = 0$ (Speed has no linear relationship with distance)

$H_1 : \beta_1 \neq 0$ (Speed has a significant linear relationship with distance)

- (g) Based on the output in (c), is the slope statistically significant at the 5% level? Why?

Yes. The p-value for speed is 1.49×10^{-12} (which is 0.00000000000149). Since this is far less than 0.05, we reject the null hypothesis.

- (h) What is the F-statistic for this model?

89.57

- (i) What is the p-value associated with the F-statistic? What does it conclude?

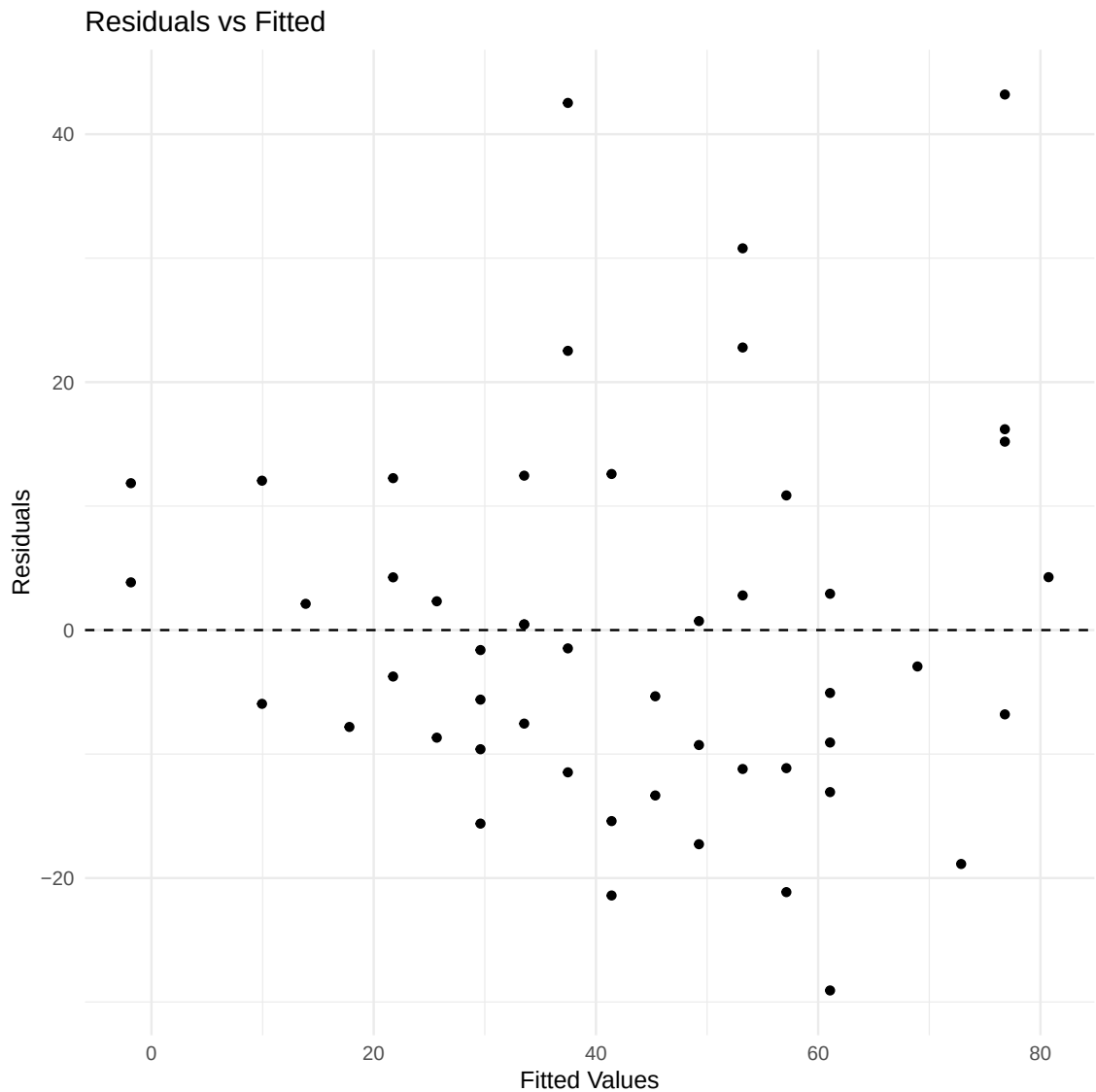
The p-value is $1.49e - 12$. This tests the Null Hypothesis that all slope coefficients are zero ($H_0 : \beta_1 = 0$). Since $p < 0.05$, the overall model is statistically significant. (Note: In Simple Linear Regression, the p-value of the F-statistic is identical to the p-value of the t-test for the slope).

(j) What is the Multiple R-squared (R^2) value, and what does it tell us?

0.6511. This means that approximately 65.11% of the variation in stopping distance (dist) is explained by the car's speed.

(k) Examine the residual diagnostics of the fitted model. Comment on whether the regression assumptions appear to be reasonable.

```
ggplot(data.frame(  
  fitted = fitted(model),  
  residuals = resid(model)  
) , aes(x = fitted, y = residuals)) +  
  geom_point() +  
  geom_hline(yintercept = 0, linetype = "dashed") +  
  labs(  
    x = "Fitted Values",  
    y = "Residuals",  
    title = "Residuals vs Fitted"  
  ) +  
  theme_minimal()
```



The residual plots show no strong pattern, and the normal Q-Q plot is approximately linear, indicating that the regression assumptions are reasonably satisfied.

- (1) From a road safety perspective, discuss whether a linear model is appropriate for predicting stopping distance at very high speeds. Justify your answer.

While the linear model fits the observed data well within the range of the dataset, extrapolating to very high speeds may be inappropriate. Physically, stopping distance is influenced by braking mechanics and reaction time, which may introduce nonlinearity at high speeds. Therefore, caution should be exercised when using this linear model beyond the observed speed range.

8.2 Multiple Linear Regression

While Simple Linear Regression models the relationship between a response variable and a single predictor, **Multiple Linear Regression** extends this to include multiple independent variables. This allows us to control for confounding variables and improve predictive accuracy.

The model takes the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

Example 8.2:

Using the built-in `mtcars` dataset, we wish to predict a car's fuel efficiency (*mpg*) based on its weight (*wt*, in 1000lbs) and gross horsepower (*hp*).

1. Fit the multiple regression model and generate the ANOVA table.

```
# Fit the model: mpg ~ wt + hp
model_multi <- lm(mpg ~ wt + hp, data = mtcars)

# Display the model summary
summary(model_multi)

##
## Call:
## lm(formula = mpg ~ wt + hp, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.941 -1.600 -0.182  1.050  5.854
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.22727    1.59879   23.285 < 2e-16 ***
## wt          -3.87783    0.63273   -6.129 1.12e-06 ***
## hp           -0.03177    0.00903   -3.519 0.00145 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.593 on 29 degrees of freedom
## Multiple R-squared:  0.8268, Adjusted R-squared:  0.8148
## F-statistic: 69.21 on 2 and 29 DF, p-value: 9.109e-12

# Generate the Analysis of Variance (ANOVA) table
anova(model_multi)

## Analysis of Variance Table
##
## Response: mpg
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wt           1  847.73   847.73 126.041 4.488e-12 ***
## hp           1   83.27    83.27  12.381 0.001451 **
## Residuals  29  195.05     6.73
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2. Write down the estimated regression equation.

$$\hat{Y} = 37.23 - 3.88X_{wt} - 0.03X_{hp}$$

3. Interpret the ANOVA table results.

- *wt* (Weight): The first row tests the significance of *wt* alone. With a p-value of 4.4910^{12} , weight is a highly significant predictor of mpg.
- *hp* (Horsepower): The second row tests the significance of adding *hp* to a model that already contains *wt*. The F-value is 12.38 with a p-value of 0.00145. Since $p < 0.05$, adding horsepower significantly improves the model even after accounting for weight.

4. Hypothesis Testing for the Overall Model

$H_0 : \beta_{wt} = \beta_{hp} = 0$ (None of the predictors have a relationship with mpg).

$H_1 : \text{At least one } \beta_j \neq 0.$

The F-statistic is 69.21 with a p-value of 9.1110612. Since the p-value is practically zero, we reject H_0 and conclude that the overall model is significant.

8.3 Further Research: Advanced Regression Techniques

(Self-Study Section)

While Simple and Multiple Linear Regression are powerful, real-world data in AI often exhibits complex, non-linear patterns or involves categorical outcomes. The following techniques are standard industry extensions that students are encouraged to explore.

1. Polynomial Regression

When the relationship between the independent variable X and the dependent variable Y is non-linear, a straight-line model may underfit the data. Polynomial regression extends the linear regression framework by including polynomial terms of the predictor variable (such as X^2 , X^3 , etc.) as additional features.

- Key concept: Instead of fitting a straight line, the model fits a smooth curve to the data points.
- Use case: Modeling growth rates, physical trajectories, or saturation-type relationships.

2. Logistic Regression

Despite its name, logistic regression is used for classification rather than predicting continuous outcomes. It models the probability that an observation belongs to a particular class (for example, *Spam* versus *Not Spam*).

- Key concept: The model uses the sigmoid function to map predictions to the interval $[0, 1]$, producing an S-shaped curve instead of an unbounded straight line.
- Use case: Predicting binary outcomes such as customer churn (Yes/No) or disease diagnosis (Positive/Negative).

3. Ridge and Lasso Regression (Regularization)

In data science, overfitting is a common issue, where a model captures noise in the training data rather than the underlying pattern, resulting in poor generalization to new data.

- Ridge regression (L2): Introduces a penalty on the squared magnitude of the coefficients, shrinking them toward zero and reducing model complexity.
- Lasso regression (L1): Applies a penalty on the absolute values of the coefficients, which can force some coefficients to be exactly zero, thereby performing automatic feature selection.
- Use case: Analyzing high-dimensional datasets (such as genomic data) or controlling overfitting in complex predictive models.
